Date: 2026-10-12
To: Editor-in-Chief, IEEE Transactions on Information Forensics and Security
Manuscript ID: T-IFS-28362-2026 (existing draft; not a new submission)
Title: Probing-Resistant Masked Hardware: Formal Security Proofs and Simulation-Based Side-Channel Verification

Dear Editor,

Following the editorial notice of 9 October 2026, the appendix previously uploaded as a separate file has been relabeled as *Supplemental Material* and is uploaded as such. The main manuscript is self-contained and is at most 13 pages in the two-column IEEE format.

Contributions.

- A characterization of the soundness gap between equipment-free value-based statistical verification and glitch-extended probing, via a cone-non-completeness condition; a first-order lint shown sound, and an explicit first-order witness showing that the gap is real.
- A masked multiplication gadget (DOM with register barrier) proven robust d-NI with perfect simulation for arbitrary d, and a first-order AES S-box built from it.
- A reproducible, equipment-free verification flow for the first-order design combining the lint, PROLEAD and gate-level TVLA/MI, calibrated with an unmasked positive control.

Scope. The empirical evaluation targets the first-order design; the gadget-level proof holds for arbitrary d, and the lint theorem is stated at first order. Limitations are discussed in the paper.

Supplemental Material. Corrected PROLEAD worst-case value, the explicit gap witness, the pre-barrier PROLEAD history, extended discussion, further verifiers and full-chip details.

Artifact. RTL, synthesis scripts, lint, PROLEAD configurations and results are provided in the artifact bundle accompanying the submission.

The manuscript is not published elsewhere or under consideration at another journal. All authors approve this submission. No conflict of interest.

Sincerely,
The Authors